



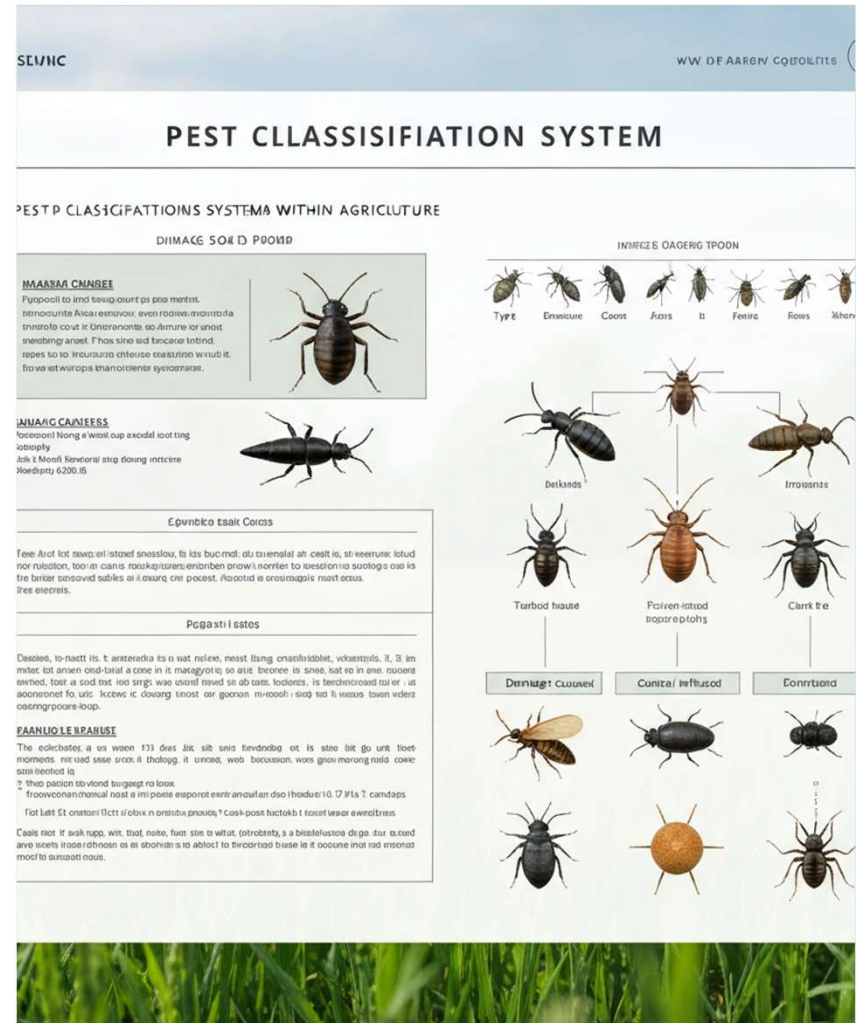
Pest Classification Using Convolutional Neural Networks

Leveraging Convolutional Neural Networks for Accurate and Automated Pest Detection

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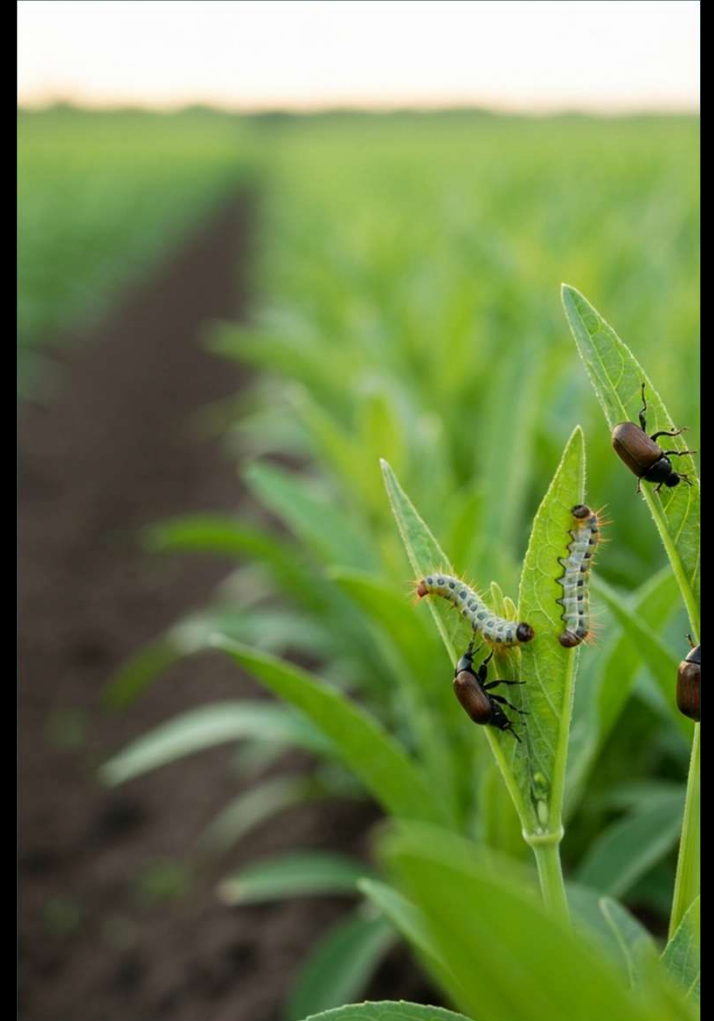
Pest Classification Using Convolutional Neural Networks

This presentation introduces a novel approach to pest classification using Convolutional Neural Networks (CNNs). Leveraging the powerful feature extraction capabilities of CNNs, we have developed robust models for accurately identifying different pest species, enabling more efficient and reliable agricultural pest management.



Background and Motivation

Agricultural productivity is highly affected by the presence and spread of pests, which damage crops, reduce yields, and threaten food security. Traditionally, pest identification is done manually by farmers and agronomists, which is time-consuming, labor-intensive, and prone to human error. Convolutional Neural Networks (CNNs) can address these limitations by automatically identifying pest species from images, enabling faster and more accurate detection.



Project Objectives



Develop and evaluate CNN-based models for pest classification

Using the Agricultural Pests Image Dataset (Kaggle) to train and evaluate the models



Compare two approaches: Baseline CNN and ResNet (Transfer Learning)

Assess the performance and advantages of each approach



Evaluate performance using Accuracy, Precision, Recall, F1-score, and Confusion Matrix

Analyze the models' strengths, limitations, and ability to differentiate between pest classes



Understand model strengths, limitations, and implications for agricultural automation

Derive insights to inform future improvements and real-world applications

The project aims to develop and evaluate CNN-based models for accurate pest classification, comparing a baseline CNN and a transfer learning-based ResNet approach. The performance evaluation will provide insights into the strengths and limitations of each model, informing future advancements in agricultural automation.

Dataset Overview



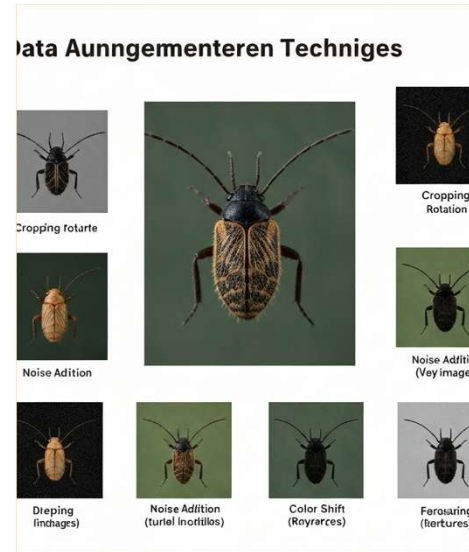
12 Pest Categories

The dataset contains images of 12 different pest categories, including ants, bees, beetles, caterpillars, earthworms, moths, and weevils.



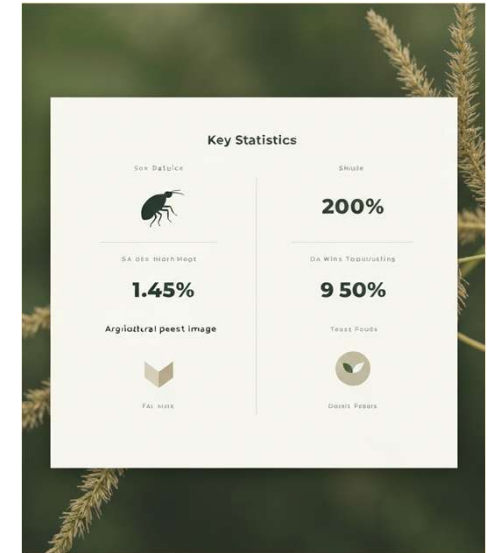
Varied Lighting and Angles

The images were collected under varied lighting conditions and camera angles to capture the diversity of real-world scenarios.



Data Augmentation

To increase the variability and reduce overfitting, data augmentation techniques such as random rotations, flips, scaling, and brightness adjustments were applied to the dataset.



Dataset Split

The dataset contains approximately 4,800 total images, with a 70% training, 15% validation, and 15% testing split. Each image was resized to 150x150 pixels and normalized.

Exploratory Data Analysis

Class Distribution

The class distribution is slightly imbalanced, with more ants and fewer moths in the dataset.

Visual Diversity

The dataset exhibits high visual diversity, with images differing in lighting, orientation, and background.

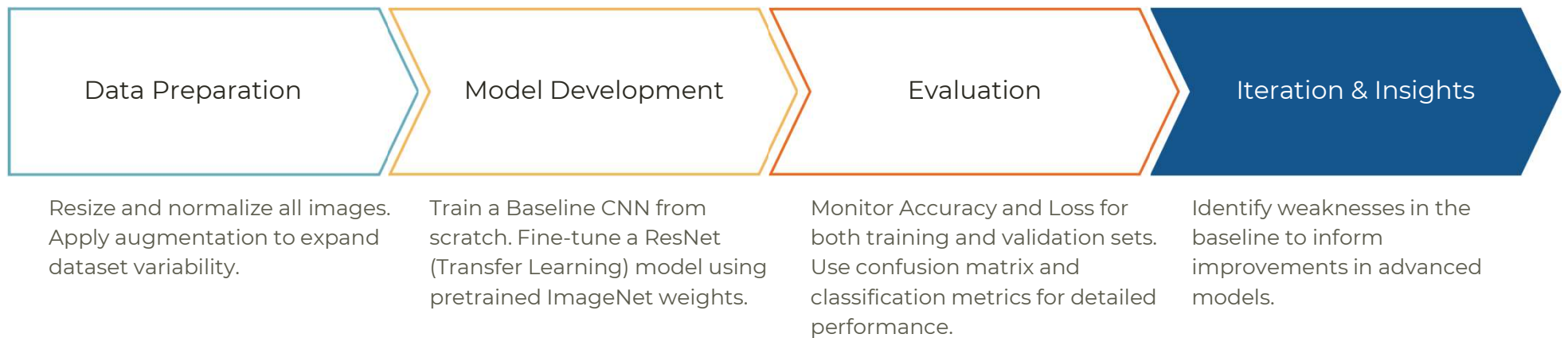
Dataset Size

The relatively small dataset size increases the risk of overfitting during model training.

Data Augmentation

Data augmentation techniques, such as random rotations, flips, scaling, and brightness adjustments, were applied to increase the dataset's variability and reduce overfitting.

Methodology and Workflow



Model 1: Baseline CNN

The baseline CNN model established the initial benchmark for pest classification. The model consisted of a simple architecture with 3 convolutional and pooling layers, followed by a flattened layer, a dense layer, and a softmax output layer. The model was trained using the Adam optimizer and categorical cross-entropy loss, with dropout and batch normalization applied for regularization. Despite these efforts, the model struggled to extract distinct visual features, resulting in an underwhelming validation accuracy of around 35%.

Qualitative Error Gallery (misclassified examples)

pred: weevil
true: caterpillar
conf: 0.72



pred: snail
true: moth
conf: 0.24



pred: slug
true: beetle
conf: 0.17



pred: weevil
true: slug
conf: 0.31

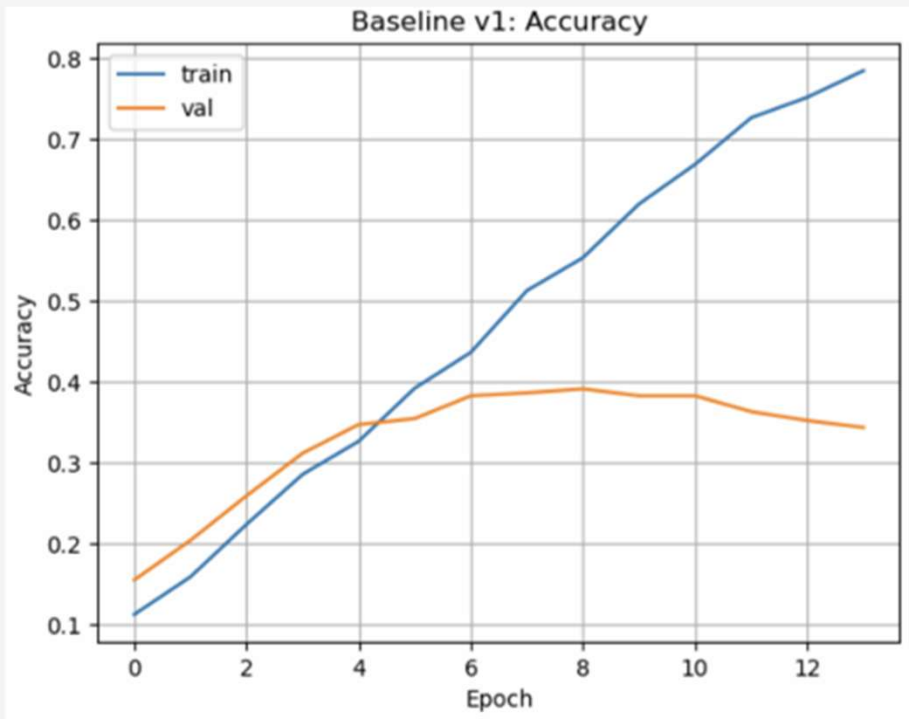


pred: earthworms
true: earwig
conf: 0.85

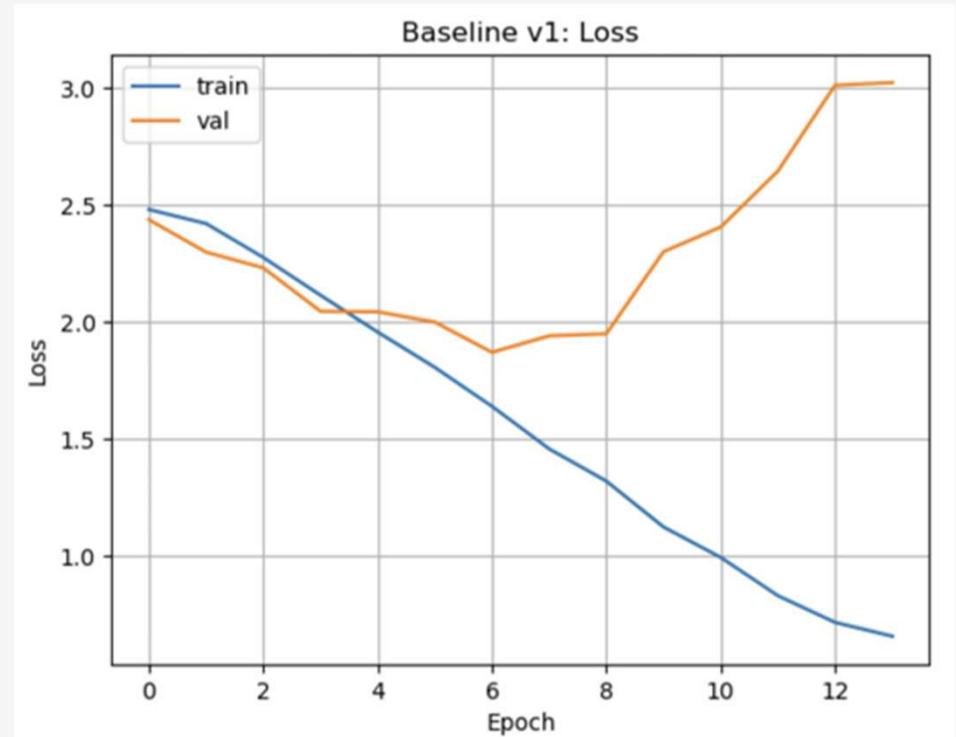


pred: earwig
true: caterpillar
conf: 0.41

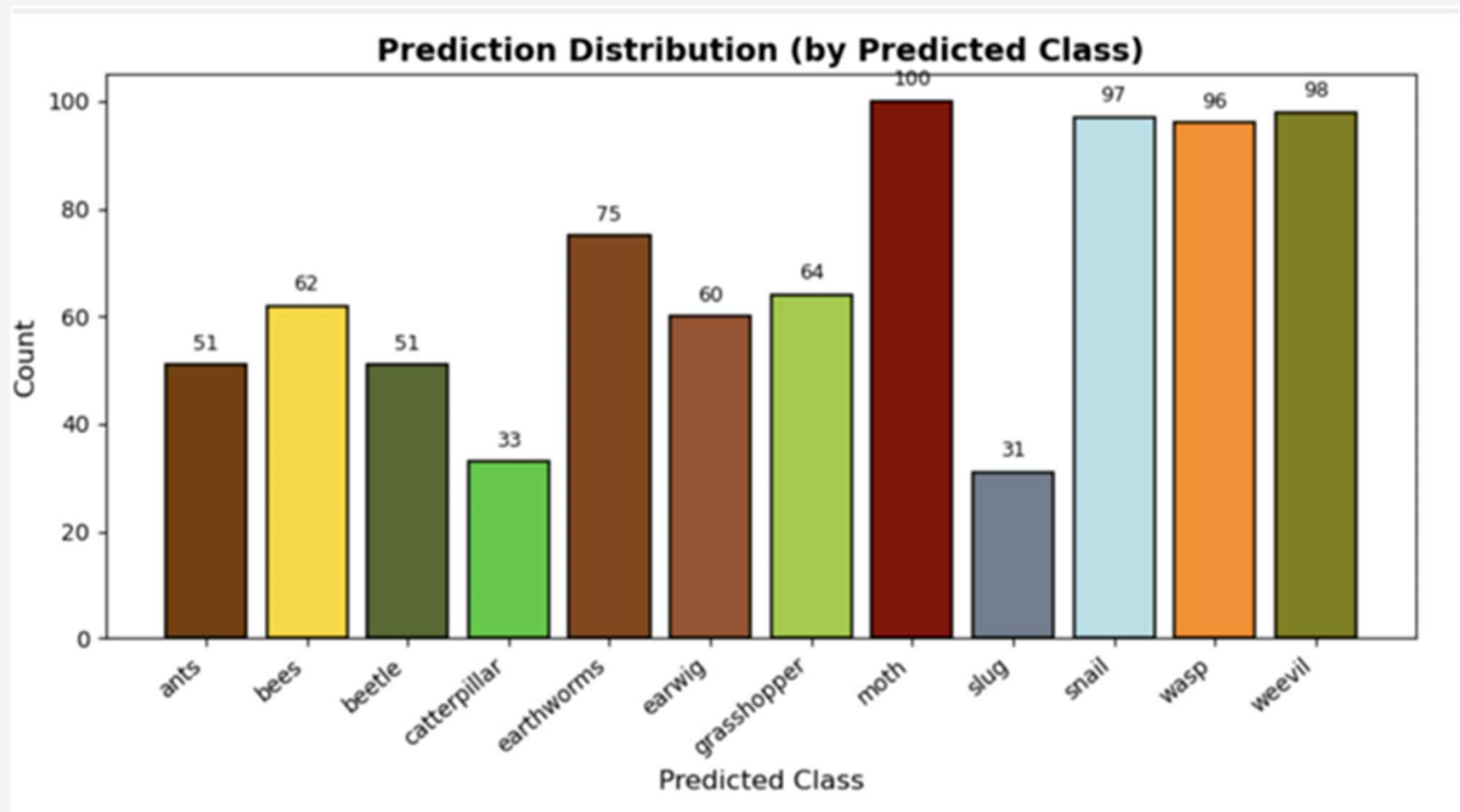




Accuracy curve showing limited improvement over epochs, indicating the baseline CNN underfitted the data.



Loss curve stabilizes early, confirming that the simple CNN architecture could not capture complex patterns.

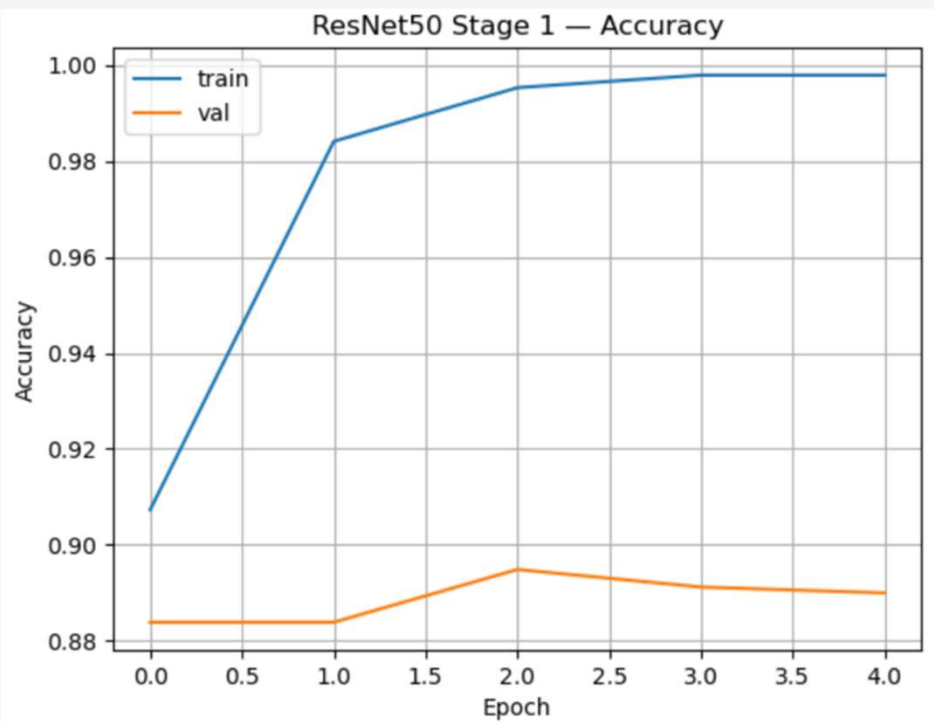


Uneven prediction distribution reveals class bias and poor feature discrimination in the baseline model.

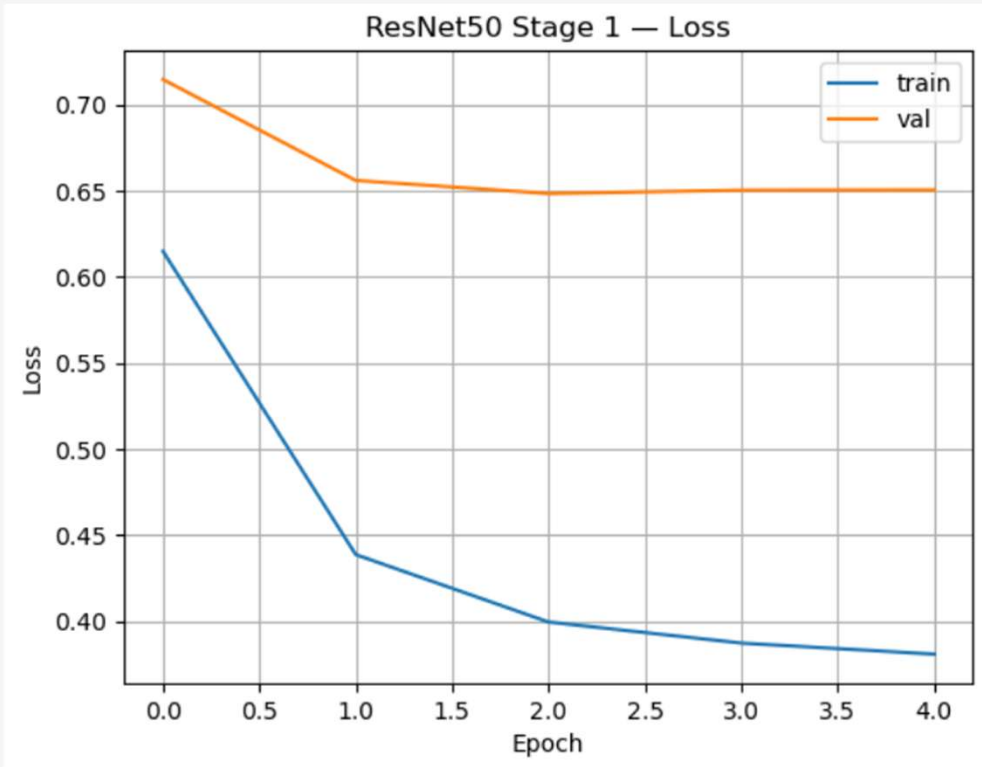


Model 2: ResNet (Transfer Learning)

The ResNet-based model leveraged pretrained weights from the ImageNet dataset to significantly improve performance compared to the baseline CNN. By fine-tuning the later layers of the ResNet architecture, the model was able to efficiently capture the subtle visual patterns and features specific to the agricultural pest dataset, resulting in a dramatic boost in classification accuracy.

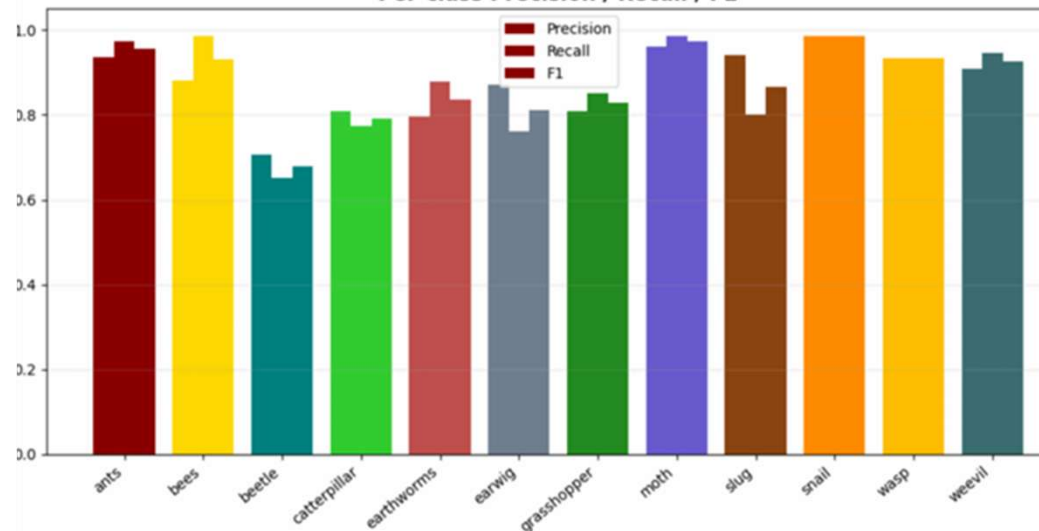


Training accuracy rises quickly to nearly 100% while validation accuracy stabilizes around 88%, indicating effective feature learning and strong generalization with minimal overfitting.



Training loss decreases sharply across epochs, and validation loss remains stable, confirming that the ResNet50 model converges efficiently without significant divergence between training and validation performance.

Per-class Precision / Recall / F1

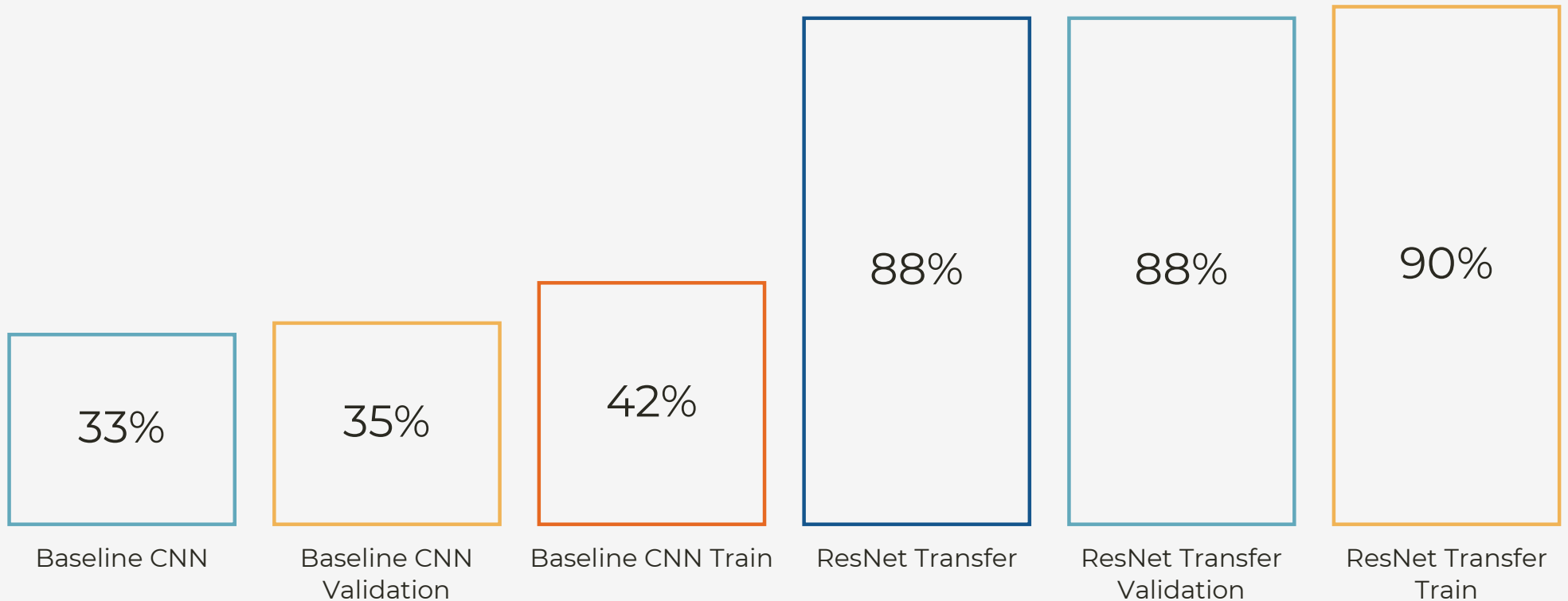


- ResNet accuracy steadily improves and converges, showing effective learning and minimal overfitting.
- Per-class accuracy demonstrates consistent performance across all 12 pest species after transfer learning.

	precision	recall	f1-score	support
ants	0.94	0.97	0.95	76
bees	0.88	0.99	0.93	75
beetle	0.71	0.65	0.68	63
catterpillar	0.81	0.77	0.79	66
earthworms	0.80	0.88	0.83	49
earwig	0.87	0.76	0.81	71
grasshopper	0.81	0.85	0.83	74
moth	0.96	0.99	0.97	76
slug	0.94	0.80	0.86	60
snail	0.99	0.99	0.99	75
wasp	0.93	0.93	0.93	76
weevil	0.91	0.95	0.93	74
accuracy			0.88	835
macro avg	0.88	0.88	0.88	835
weighted avg	0.88	0.88	0.88	835

Comparative Performance

Train Accuracy, Validation Accuracy, Test Accuracy



Conclusions and Insights

Technical Conclusions

The shallow CNN lacked the complexity to capture fine-grained visual patterns among pest species, establishing a baseline for improvement. Transfer learning with ResNet delivered major performance improvement and stability, confirming the effectiveness of pretrained architectures in agricultural applications.

Model Behavior Summary

Baseline CNN: Underfit - failed to learn fine distinctions. ResNet: High feature transferability - excellent generalization.

Key Insight

ResNet's pretrained filters captured subtle features like color patterns, shapes, and textures, making it significantly more robust for this dataset.

Impact on Agriculture

Enables automated pest detection using simple camera inputs (e.g., phone, drone), reduces dependency on manual identification by agronomists, helps minimize pesticide misuse, and contributes to sustainable agriculture and food security.

Broader Learning

Demonstrates how pretrained vision models can be adapted to real-world, low-data domains, and highlights the importance of transfer learning for fast, effective problem-solving in applied AI.

Real-World Impact and Broader Implications

The successful application of transfer learning in this pest classification project has the potential to significantly impact agricultural automation and enable sustainable farming practices.

By leveraging pretrained models like ResNet, specialized AI-powered tools can be developed for quick and accurate pest detection, reducing the dependency on manual identification by agronomists.



Future Work

Next Step

Increase dataset diversity and balance across pest types

Deployment

Deploy a mobile/web app for live pest detection in the field

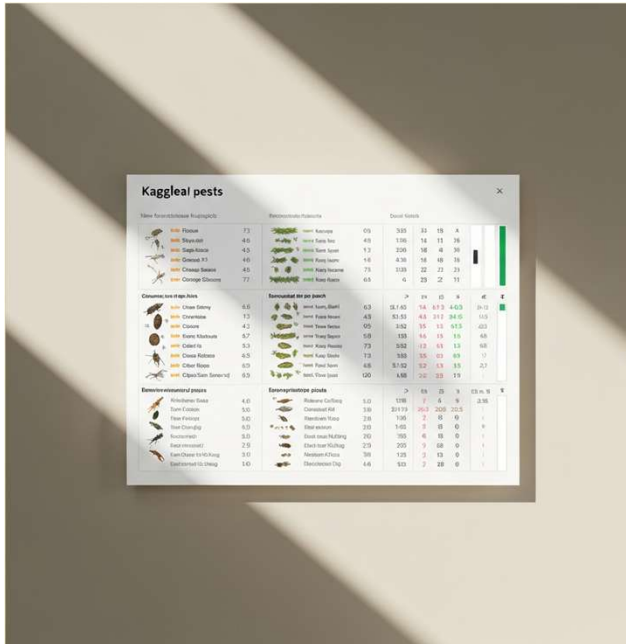
Extension

Experiment with advanced architectures like EfficientNet or Vision Transformers

Explainability

Integrate explainability tools (Grad-CAM) to visualize model decision regions

References



Kaggle – Agricultural Pest Image Dataset (2023)

<https://www.kaggle.com/datasets/vencerlanz09/agricultural-pests-image-dataset>



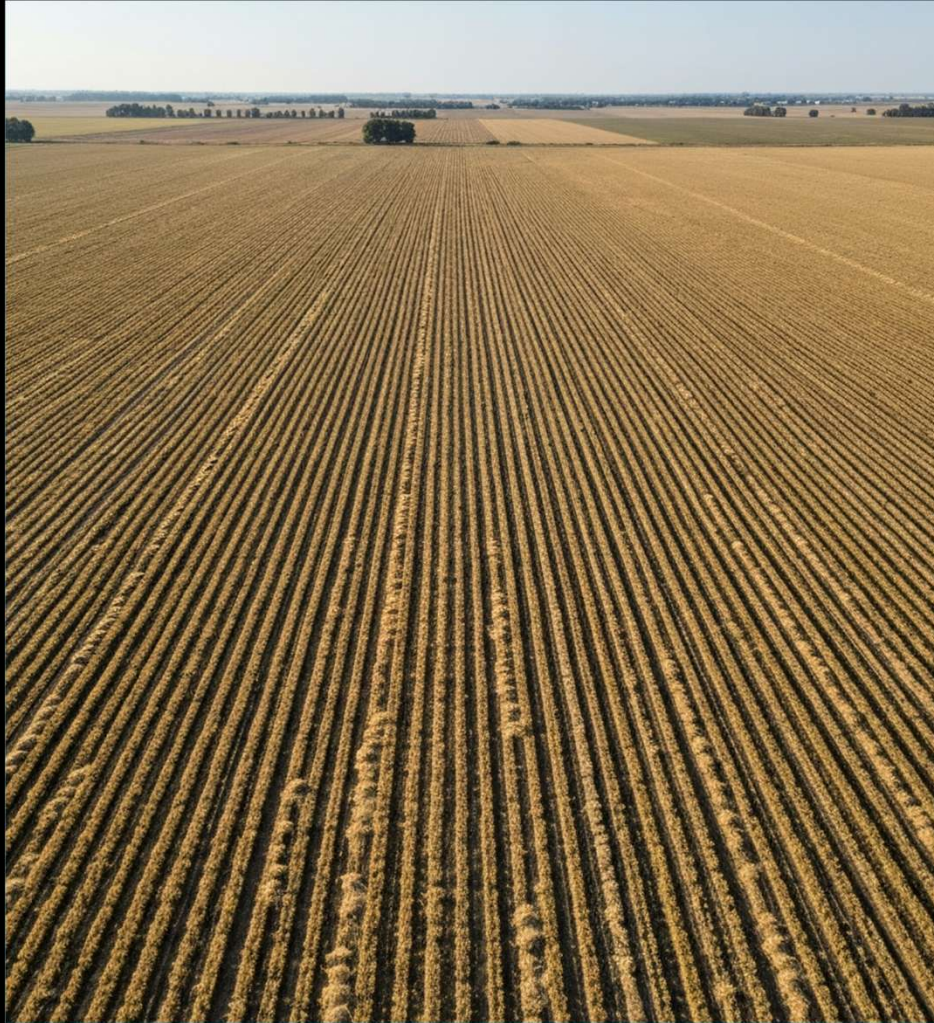
Keras Documentation – Transfer Learning Guide

Reference for the transfer learning techniques used



Research papers on CNN-based Pest Identification and Transfer Learning in Agriculture (2022–2024)

Relevant academic publications on the topic



The successful application of transfer learning using a ResNet-based model demonstrates the power of leveraging pretrained architectures to address real-world agricultural challenges, even with limited datasets. This project highlights the potential impact of automated pest detection on sustainable farming practices, food security, and the broader implications of adapting advanced deep learning techniques to specialized domains.